Finale file formats

A reader's specification for `.mus` and `.musx`

Reconstructed by the finale-file-parser project. All hex dumps are synthetic.

Contents

1. Scope and provenance

This document describes the binary layout of Finale's `.mus` and `.musx` files as reconstructed by the `finale-file-parser` project. Finale was discontinued in 2024 and its format was never published; everything here comes from analyzing a curated corpus, from two Coda documents[1][2] vendored into the repository, and from prior community research.[3]

It is written for someone implementing a reader. Every structure is given as a C-style declaration, a field table, and a hex dump.

> ***Confidence.*** *Each structure below is annotated with what it was verified against. Where this project's findings contradict Coda's own documentation, that is stated explicitly rather than quietly resolved — there is one such case, the note alteration encoding in §7.2.*

No corpus bytes appear in this document

Every hex dump is **synthetic**, constructed by the generator that produced this PDF. The test corpus is licensed third-party music and its bytes are not reproducible here. Synthetic data also lets each dump isolate the field under discussion instead of burying it in a real record.

> **This document is generated from the parser.** Offsets, sizes and constants are imported from the reading code rather than retyped, so a layout here cannot silently drift from the implementation that reads it.

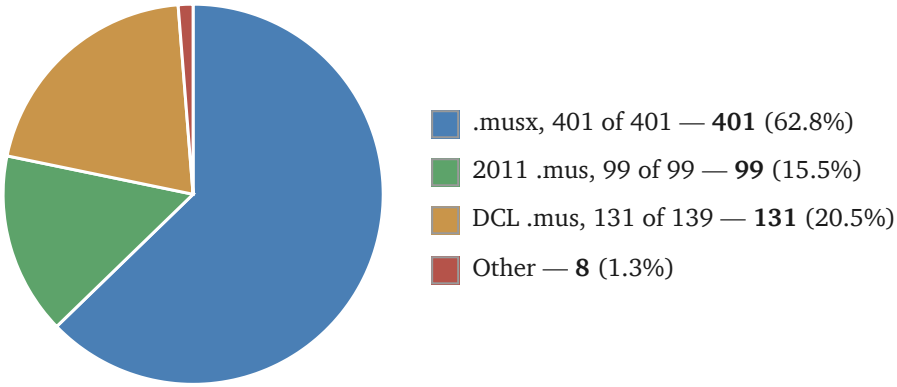
2. Which formats are supported

Three containers, spanning roughly two decades of Finale releases.

CONTAINER	FINALE ERA	PAYLOAD CODEC	BYTE ORDER	RECORD ENCODING
`.mus` (DCL)	2001–2005	PKWARE DCL (“implode”)	writing platform's	fixed 16-byte ETF rows
`.mus` (2011)	2011–2012	zlib (chained streams)	little-endian	self-identifying variable-length
`.musx`	2012–2024	ZIP + XOR cipher, then zlib	little-endian	EnigmaXML elements

Corpus coverage

The figures the parser is measured against, over 639 documents. 631 build a score.



The other eight files

Six empty scores

These files appear to be empty. They carry staves and measures, but no notes. The reader refuses them rather than returning an empty score, because an empty result is indistinguishable from a parse that went wrong.

One mirror

Mirror is Coda's own term — their 1996 documentation[4] warns that “mirrors and voice 2 create complications”, and Finale shipped a Mirror Tool for creating them.

A mirror is a staff that *displays another staff's music instead of holding its own copy*. An engraver reaches for one when two parts play the same thing — a doubled line, a cue, a piano reduction of what the winds are doing. Rather than duplicating the notes, the second staff is pointed at the first, so editing the original changes both.

Stored, this means exactly what it sounds like: there is one set of entries, and two `gfhold` records name the same entry span. Nothing marks either as the copy. To a reader walking the frame chain, the same entries simply turn up twice, in two different places in the score.

Five DCL documents in the corpus contain mirrors; four read fine, because their mirrored spans are named once. The one that fails is the only document where *two* `gfhold` records name the same entry span, which would require a single entry to exist in two places at once. The intermediate representation gives an entry exactly one location, so supporting this is a design change rather than a decoding problem.

One with a measure count mismatch

A file has 36 measures declared in the `measSpec` records, but measures as high as 111 appear in the `gfhold` records. The measures being referenced are not in the file, so most of the score cannot be reconstructed. Its sibling in the same folder carries a `_Temp` suffix, which suggests a working file rather than a finished one, though what truncated it is not recorded anywhere in the format.

> **Not covered.** Finale versions between 2005 and 2011, and before 2001, are absent from the corpus and therefore absent from this document. Nothing here should be assumed to hold for them.

3. Byte order, compression, and encryption

Byte order is the writing machine's

For DCL-era `.mus` files, integers are written in the byte order of the machine that saved the file: **big-endian on Mac, little-endian on Windows**. This governs the pool records and every field inside them.

It is *detected, not assumed*, and the first pool record makes that possible. Its `kind` field is always 15, and 15 in a 16-bit field is `0f 00` read little-endian but `00 0f` read big-endian. So a reader tries one order, and if the first two bytes do not give 15 it tries the other: the wrong order yields 3,840, which is not a pool kind. One field, read two ways, decides the whole file.

The check that the choice was right is that the **chain walks to the last byte exactly**. Each pool record's `length` says how far the next one begins, so a reader adds lengths from `0x200` and lands on record after record. If the byte order were wrong the lengths would be wrong, and the walk would overshoot the end of the file or stop short of it. Landing precisely on the final byte, with no gap and no overrun, is strong evidence that every length was read correctly. All 139 DCL documents in the corpus do this: 102 little-endian, 37 big-endian.

> All files in the 2011 era are **little-endian**, since this is after Apple switched to Intel-based Macs.

Compression

DCL era (2001–2005)
: A chain of PKWARE DCL (“implode”) records beginning at `0x200`. No Python standard-library module reads this format. This project **wrote its own decoder**, an independent Python port of Mark Adler's `blast.c` from zlib's `contrib/blast`, whose correctness is pinned by that implementation's own published test vector.

2011 era
: A chain of raw zlib streams, the first located by scanning for the `78 9c` header. The standard library reads these.

`.musx`
: A ZIP archive. The member `score.dat` is XOR-obfuscated, and decodes to zlib-compressed EnigmaXML.

For scale rather than as a specification: a DCL payload decompresses to about 3.3–4.5 times its stored size, and a 2011 payload to about 5.9–8.6 times. Decoded payloads in the corpus run from 32 KB to 683 KB.

Reading a file you did not write

Every input is hostile until parsed. A specification that describes only well-formed files leaves a reader open to three distinct attacks, which need three distinct defenses — conflating them is how one gets missed.

Decompression bombs

A small file can declare an enormous decompressed size, exhausting memory before anything is validated. Neither DCL nor zlib bounds its own output. This parser **rejects any score whose decoded payload exceeds 64 MiB**, counted across the whole pool chain rather than per stream, so a chain of individually modest pools cannot add up to an unbounded total. The largest real payload in the corpus is 683 KB, so the limit sits about 90 times above anything legitimate.

XML entity attacks

A `.musx` carries XML, and XML has billion-laughs entity expansion and external-entity (XXE) retrieval built into the standard. All XML in this project is parsed with **`defusedxml`**, which disables entity expansion and external references. This is the project's only runtime dependency, and it exists for this reason alone.

Malformed offsets and lengths

Every offset and length in this document is read *from the file* and so may be a lie. A record can declare a length running past the end of its pool, a frame can name entries that do not exist, and a walk can be steered into an infinite loop. Each such value is bounds-checked before use, and a file that fails a check raises a clear error naming what was wrong — never a crash, a hang, or a silent truncation.

The `score.dat` obfuscation

`score.dat` is XOR-ed with a keystream from a BSD `rand()` linear congruential generator, whose state advances as $\texttt{state} = \texttt{state} \times \texttt{0x41c64e6d} + \texttt{0x3039}$ (mod 2^{32}), starting from the fixed seed `0x28006d45`. The generator is restarted from that same seed at every 128 KiB boundary, so the keystream is one 128 KiB block repeated end to end for the whole file.

> **This is not a block cipher, and it is not encryption.** There is no mode of operation in the usual sense — no ECB, CBC or CTR — because there is no block cipher and no key. It is a *stream cipher* whose keystream is generated independently of the data, which makes it structurally like OFB or CTR mode; but the seed is a constant compiled into the application, so the keystream is **identical in every file ever written**. That makes it obfuscation, not confidentiality: anyone with one plaintext and its ciphertext recovers the keystream, and the repetition every 128 KiB is the classic many-time-pad weakness on top. Treat `score.dat` as plainly readable.

> *These cipher parameters were not discovered by this project. They come from denigma (MIT), whose source credits Deguerre.*

4. The `.mus` file header

Both `.mus` eras share a header. Its first `0xA0` bytes carry the version banner and two provenance stamps.

A synthetic .mus header. The banner is the primary version evidence; the two provenance stamps say which application and platform wrote the file.

```
// byte order: little-endian (this example)
struct MusFileHeader {
    char[64]  banner;                   // +0x20   NUL-terminated; tail of a previ...
    uint32    created.date;             // +0x66   creation date stamp
    char[4]   created.app;              // +0x70   application tag, e.g. FIN
    char[4]   created.platform;         // +0x74   MAC or WIN
    uint32    modified.date;            // +0x8c   last-modified date stamp
    char[4]   modified.app;             // +0x96   application tag
    char[4]   modified.platform;        // +0x9a   MAC or WIN
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x20–0x5f	64	char[64]	banner	NUL-terminated; tail of a previous longer banner may follow
	0x66–0x69	4	uint32	created.date	creation date stamp
	0x70–0x73	4	char[4]	created.app	application tag, e.g. FIN
	0x74–0x77	4	char[4]	created.platform	MAC or WIN
	0x8c–0x8f	4	uint32	modified.date	last-modified date stamp
	0x96–0x99	4	char[4]	modified.app	application tag
	0x9a–0x9d	4	char[4]	modified.platform	MAC or WIN

Example bytes (160 shown), tinted to match the table above.

```
0000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0010  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0020  46 69 6e 61 6c 65 28 52 29 20 32 30 30 35 20 66  Finale(R) 2005 f
0030  6f 72 20 57 69 6e 64 6f 77 73 00 00 00 00 00 00  or Windows......
0040  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0050  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0060  00 00 00 00 00 00 80 3a 09 00 00 00 00 00 00 00  .......:........
0070  46 49 4e 00 57 49 4e 00 00 00 00 00 00 00 00 00  FIN.WIN.........
0080  00 00 00 00 00 00 00 00 00 00 00 00 10 3b 09 00  .............;..
0090  00 00 00 00 00 00 46 49 4e 00 57 49 4e 00 00 00  ......FIN.WIN...
```

> The banner field is fixed-size and is *not* zero-filled when Finale rewrites it, so a shorter banner can leave the tail of a previous, longer one behind. Everything from the first NUL onward must be discarded.

> A stamp is all-or-nothing: an implausible date or an empty application tag means the whole stamp is untrustworthy, because a reader cannot tell which half of a partial stamp to believe.

The banner is the primary version evidence. A regular expression of the form `Finale\(R\)\s+(\d{4})` against the banner text yields the year. An unrecognized banner should leave the year unset with the raw text preserved, rather than failing — an unknown variant stays inspectable that way.

5. Container: how the payload is found

A `.mus` payload is **not one blob**. It is a handful of compressed *pools*, laid end to end.

POOL	KIND	HOLDS
others	15	most record types, keyed by one cmper
details	16	records keyed by a pair of cmpers
entries	17	the notes themselves
text	18	human-readable strings

DCL era: a labeled chain

Records run from `0x200` to the last byte of the file, with no gaps. The DCL era **labels** its pools; the zlib era does not.

A DCL-era pool record, little-endian (Windows-written). Records lie end to end from 0x200 to the last byte of the file, with no gaps.

```
// byte order: little-endian
struct DclPoolRecord {
    uint16   kind;                      // +0x0    15 others, 16 details, 17 entri...
    uint32   length;                    // +0x2    whole record, this 10-byte head...
    uint32   checksum;                  // +0x6    omitted entirely when length == 6
    uint8[]  stream;                    // +0xa    PKWARE DCL data, length - 10 bytes
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	kind	15 others, 16 details, 17 entries, 18 text
	0x02–0x05	4	uint32	length	whole record, this 10-byte header included
	0x06–0x09	4	uint32	checksum	omitted entirely when length == 6
	0x0a–0x10	7	uint8[]	stream	PKWARE DCL data, length - 10 bytes

Example bytes (17 shown), tinted to match the table above.

```
0000  0f 00 11 00 00 00 9c 21 44 00 00 04 0e c5 b3 9a  .......!D.......
0010  2f                                               /
```

> **length counts its own header.** The chain is walked by adding length to the current position. This is also why a fixed 0x20A works as "where the first stream starts": 0x200 plus the ten-byte header.

> A length of exactly 6 means the pool is **empty** — kind and length and nothing else, no checksum and no stream.

The identical record written big-endian (Mac). Byte order is detected from this first record rather than assumed: its kind is always 15, which reads as 15 only one way round.

```
// byte order: big-endian
struct DclPoolRecord {
    uint16   kind;                      // +0x0    15 — only 15 one way round
    uint32   length;                    // +0x2
    uint32   checksum;                  // +0x6
    uint8[]  stream;                    // +0xa
};
```

OFFSET	SIZE	TYPE	FIELD	MEANING
0x00–0x01	2	uint16	kind	15 — only 15 one way round
0x02–0x05	4	uint32	length	
0x06–0x09	4	uint32	checksum	
0x0a–0x10	7	uint8[]	stream	

Example bytes (17 shown), tinted to match the table above.

```
0000  00 0f 00 00 00 11 00 44 21 9c 00 04 0e c5 b3 9a  .......D!.......
0010  2f                                               /
```

2011 era: an unlabeled chain

The first zlib stream is found by scanning for the `78 9c` header. Streams follow one another; the pools are identified by *order*, not by a label, which is why a reader must know the sequence rather than read it.

6. Records: framing and addressing

What a record is

A record is **a row in a table, not an object in a tree**. Almost every wrong intuition about these formats comes from expecting a document tree, so it is worth stating plainly before any bytes.

A record is three things:

A tag
: What kind of thing this is — `measSpec`, `gfhold`. Think table name.

One or more keys
: What it is *about*. A `measSpec` keyed 7 is measure 7; a `gfhold` keyed (3, 7) is staff 3, measure 7. Think primary key.

A payload
: Bytes whose meaning depends entirely on the tag. Think columns. There is no self-description inside a payload: without the tag, the bytes mean nothing.

Three consequences matter to anyone writing a reader.

There are no pointers. Nothing holds a file offset. A `gfhold` does not point at a `frameSpec`; it contains the number 41, and somewhere there is a `frameSpec` whose key is 41. Resolution is a lookup by key equality — a join, not a dereference. Even the entry pool's `next` and `prev`, which look like a linked list, are entry *numbers*. The physical order of every record in a file could be shuffled without losing anything.

Records do not nest. Composition happens by reference. The music of a measure is not inside the measure's record; it is reached by a chain of key lookups.

A logical record is not always a physical one. In the 2011 era they are one to one — each record carries its own length. In the DCL era a record is a fixed 16-byte row, and anything larger continues into further rows under the same tag and key. ETF calls each row an *incidence*. So a record is the concatenation of its rows, and fields are addressed by offset into that concatenation, not into any one row.

The hierarchy, and where containment actually happens

"A pool contains records, and records contain entries" is half right. Pools do contain records. But an entry **is** a record: it lives in its own pool, keyed by entry number, a peer of `measSpec` rather than a child of anything.

What misleads is `frameSpec`, which looks like containment and is not. It holds `startEntry` and `endEntry` — two integers naming a range of keys. Delete the `frameSpec` and its entries are still there, orphaned but intact.

```
file
└── pools                    others · details · entries · text
    └── records              a pool is a flat run of them
        └── (entries pool)
            └── entry        a record: one chord or rest at one point in time
                └── notes    6-byte note records — real containment
```

Notes are the one genuine nesting in the format. An entry's payload holds its own `noteCount` and the note records themselves; they have no independent key and cannot be addressed from outside. Everything else is reference by key.

> In one line: **pools contain records; records reference each other by key; only entries contain anything, and what they contain is notes.**

2011 era: self-identifying records

A 2011-era record: tag 176 (measSpec), key 1, score part. One record occupies 10 + length + 4 bytes.

```
// byte order: little-endian
struct MusRecord2011 {
    uint16   tag;                    // +0x0    record type; .musx names the sa...
    uint16   cmper;                  // +0x2    the (n) in an ETF ^XX(n) — the...
    uint16   part;                   // +0x4    0 for the score, then 1, 2, ......
    uint32   length;                 // +0x6    payload size in bytes
    uint8[]  payload;                // +0xa    length bytes
    uint32   trailer;                // +0x16   the length again
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	tag	record type; .musx names the same types
	0x02–0x03	2	uint16	cmper	the (n) in an ETF ^XX(n) — the record's key
	0x04–0x05	2	uint16	part	0 for the score, then 1, 2, … per linked part
	0x06–0x09	4	uint32	length	payload size in bytes
	0x0a–0x15	12	uint8[]	payload	length bytes
	0x16–0x19	4	uint32	trailer	the length again

Example bytes (26 shown), tinted to match the table above.

```
0000  b0 00 01 00 00 00 0c 00 00 00 60 00 58 02 04 00  ..........`.X...
0010  01 00 00 00 00 00 0c 00 00 00                    ..........
```

> These records are **self-identifying**: each carries its own key, so nothing outside the record is needed to address it — no directory, no key array, no positional convention.

> Records of one tag sit together in a section; sections may be separated by two-byte zero padding, which a walk must skip.

> Why this took so long to find: earlier attempts anchored on a payload field located by searching for a value already known from a paired .musx, then read *forward*. The key sits ten bytes *behind* that anchor, so it was never in view.

DCL era: fixed ETF rows

The same information, encoded completely differently. Where a 2011 record carries its own length, a DCL row is always 16 bytes and a long record simply continues into the next row.

A 2001–2005 others row: fixed 16 bytes, ETF's two-character tag.

```
// byte order: writing platform's
struct DclOthersRow {
    uint16     cmper;                   // +0x0    the (n) in an ETF ^XX(n)
    uint16     tag;                     // +0x2    two characters stored as a u16
    uint8[12]  data;                    // +0x4    ETF's 6 twobyte values (or 3 fo...
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	cmper	the (n) in an ETF ^XX(n)
	0x02–0x03	2	uint16	tag	two characters stored as a u16
	0x04–0x0f	12	uint8[12]	data	ETF's 6 twobyte values (or 3 fourbytes)

Example bytes (16 shown), tinted to match the table above.

```
0000  01 00 53 4d 60 00 58 02 04 00 01 00 00 00 00 00   ..SM`.X.........
```

> **The tag is a u16, not two bytes.** On a little-endian file its characters come out reversed: ^MS is written SM, ^&a is written a&. Reading the pair verbatim finds no known tag in 102 of 139 corpus documents, which is how this was noticed.

> A record too big for one row **runs on into further rows** under the same tag and key. ETF calls each row an *incidence*. A record's payload is its rows' data concatenated in file order.

A details row. Two keys rather than one — details are addressed by a pair, which is how a record hangs off a (staff, measure) intersection.

```
// byte order: writing platform's
struct DclDetailsRow {
    uint16     cmper1;                  // +0x0    first key
    uint16     cmper2;                  // +0x2    second key
    uint16     tag;                     // +0x4    two characters as a u16
    uint8[10]  data;                    // +0x6    ETF's 5 twobytes
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	cmper1	first key
	0x02–0x03	2	uint16	cmper2	second key
	0x04–0x05	2	uint16	tag	two characters as a u16
	0x06–0x0f	10	uint8[10]	data	ETF's 5 twobytes

Example bytes (16 shown), tinted to match the table above.

```
0000  01 00 03 00 46 47 01 00 00 00 00 00 00 00 00 00  ....FG..........
```

How items point to each other

There is no pointer arithmetic anywhere in these formats. Addressing is by **key**, and the keys are small integers.

`cmper` ("comparator")
: A record's key — the `(n)` in ETF's `^XX(n)` notation. For a `measSpec` it is the measure number; for a `staffSpec`, the staff.

`cmper1`, `cmper2`
: Details records are keyed by a *pair*, which is how a record hangs off an intersection — a `gfhold`, for instance, at (staff, measure).

`inci` ("incidence")
: When several records share a key, the incidence distinguishes them. In the DCL era it is also how one logical record spills across several 16-byte rows.

`part`
: 0 for the score; 1, 2, … for each linked part. A part's record overrides the score's for that part only.

Finding the music: the frame chain

This is the central traversal, and it is worth stating as a sequence:

1. A `gfhold` ("frame hold") is keyed by (staff, measure) and names up to four **frames**, one per layer.
2. Each frame id keys a `frameSpec`, which gives a `startEntry` and an `endEntry`.
3. Those are entry numbers into the entry pool. The entries between them, inclusive, are that layer's music for that measure.
4. Each entry carries its own `next` pointer, so the run can also be walked as a linked list.

> An entry reached by more than one frame is a corrupt document, not a shared voice. A reader should refuse it rather than emit the notes twice.

Known record tags

One vocabulary in three spellings: ETF's two-character tags, the 2011 era's numbers, and EnigmaXML's symbolic names. The three tiers below are kept apart on purpose — a payload-confirmed identification and a key-sequence guess are not the same claim.

Tier A — decoded and payload-confirmed

Parsed by this project, with fields verified against paired `.musx` documents.

TAG	ETF	NAME	POOL	DESCRIPTION
109		`clefOptions`	others	Clef definition table, read as a strided array
121		`articDef`	others	An articulation's definition; charMain is the glyph
146	`^FR`	`frameSpec`	others	Entry range for one layer of one measure
159		`instUsed`	others	The staves the score lays out, in layout order
176	`^MS`	`measSpec`	others	Per measure: width, key, beats, divbeat, barline
203		`repeatBack`	others	Backward repeat barline, keyed by measure
204		`repeatEndingStart`	others	Ending bracket, keyed by measure
206		`repeatPassList`	others	Which passes an ending is taken on
231	`^IS`	`staffSpec`	others	Per staff; transposition at +0x14
1009		`articAssign`	details	The articulation on an entry; names an articDef
1044	`^GF`	`gfhold`	details	(staff, measure) to clef and up to four frames
1057		`staffGroup`	details	Brace or bracket over a run of staves
1072		`tupletDef`	details	Keyed by entry, not by (staff, measure)
1108		`lyrDataVerse`	details	Verse lyrics
—	`^eE`	`entry`	entries	The notes; fixed 38-byte slots

Tier B — named by key-sequence matching only

Identified by matching each tag's `(cmper, part)` sequence against the `.musx` `others` pool. **Treat these as leads, not facts**: a tag whose record count collides with another's could be mismatched, and none has been confirmed by payload content. “Docs” is how many corpus documents agreed.

TAG	NAME	DOCS
124	`channelPlayData`	80
126	`chordSuffixPlay`	85
131	`drumLibName`	86
134	`durAllot`	91
136	`execShape`	85
140	`fretboardSymbol`	91
144	`fontName`	5
147	`lockMeas`	81
149	`fretInst`	90
163	`layerAtts`	85
165	`metaArtic`	80
168	`metaDynam`	81

TAG	NAME	DOCS
169	`metaKeySig`	80
170	`metaRepeat`	81
171	`metaShape`	80
172	`metaStaffStyle`	80
173	`metaTimeSig`	81
235	`shapeExprDef`	10
242	`textExpressionEnclosure`	14
315	`volumeValue`	14

> `fontName` (5 documents) and `shapeExprDef` (10) rest on so few agreements that they are better read as guesses.

Tier C — observed but unidentified

189 distinct tags appear in the 2011 `others` pool alone, of which the tiers above account for roughly thirty. The rest are read, keyed and carried, but their meaning is unknown. The most frequent, by record count across the corpus:

others
: 213 and 215 (25,576 each), 140 (18,624), 214 (12,681), 125 (12,612), 148 (11,887), 126 (11,208), 183 (10,114), 192 and 241 (7,353 each), 217 (6,638)

details
: 1043 (166,524), 1064 (8,162), 1063 (2,162), 1066 (1,483), 1034 (1,164), 1060 (1,156)

DCL (ETF)
: IV (15,274), IK (15,024), SD (13,838), sL and sb (13,802 each), FB (8,448), IX (7,415), DT (4,429), TX (4,068); and in details DF (74,730), fb (63,324), DN (18,647)

> **Two hints from the counts.** Several pairs share an exact record count — 213/215, 192/241, and `sL`/`sb` — which across a whole corpus usually means a definition-and-assignment couple like `articDef`/`articAssign`. And details tag 1043, at 166,524 records, is roughly sixteen times more common than `gfhold`, which suggests something per entry or per note rather than per measure.

> ***What this means for scope.*** *The tags decoded here are not the common ones; they are the ones on the path from file to notes. Everything above is largely layout, spacing, fonts, page format and text blocks. By tag count this specification covers under a fifth of the vocabulary, and by record count rather less — it is a specification for extracting the music, not a complete description of the file.*

7. The entry pool

The notes. This is the one structure both `.mus` eras share byte-for-byte, differing only in integer byte order.

An entry's first slot. The pool is a flat array of fixed 38-byte slots, each tagged with the entry it belongs to.

```
// byte order: little-endian
struct EntrySlotFirst {
    uint32   entnum;                    // +0x0    the (n) in ETF ^eE(n)
    uint16   slot;                      // +0x4    0 for an entry's first slot, th...
    uint32   prev;                      // +0x6    previous entry number, 0 if none
    uint32   next;                      // +0xa    next entry number, 0 if none
    uint16   dura;                      // +0xe    written duration in EDU; 1024 =...
    int16    pos;                       // +0x10   manual positioning in EVPU, signed
    uint32   flags;                     // +0x12   SETBIT 0x80000000 always; NOTEB...
    uint16   extflags;                  // +0x16   extended flags
    uint16   noteCount;                 // +0x18   how many note records follow, a...
    NoteRec  note[0];                   // +0x1a   TCD (2) + note flag (4)
    NoteRec  note[1];                   // +0x20   the first slot holds two
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x03	4	uint32	entnum	the (n) in ETF ^eE(n)
	0x04–0x05	2	uint16	slot	0 for an entry's first slot, then 1, 2, ...
	0x06–0x09	4	uint32	prev	previous entry number, 0 if none
	0x0a–0x0d	4	uint32	next	next entry number, 0 if none
	0x0e–0x0f	2	uint16	dura	written duration in EDU; 1024 = quarter note
	0x10–0x11	2	int16	pos	manual positioning in EVPU, signed
	0x12–0x15	4	uint32	flags	SETBIT 0x80000000 always; NOTEBIT 0x40000000
	0x16–0x17	2	uint16	extflags	extended flags
	0x18–0x19	2	uint16	noteCount	how many note records follow, across slots
	0x1a–0x1f	6	NoteRec	note[0]	TCD (2) + note flag (4)
	0x20–0x25	6	NoteRec	note[1]	the first slot holds two

Example bytes (38 shown), tinted to match the table above.

```
0000  65 00 00 00 00 00 64 00 00 00 66 00 00 00 00 04   e.....d...f.....
0010  00 00 00 00 00 c0 00 00 02 00 c0 03 00 00 00 00   ................
0020  01 04 00 00 00 00                                 ......
```

> Continuation slots (index > 0) carry only notes, 5 per slot, from offset 6. An entry may carry up to 12 notes.

> **Both eras share this layout.** What differs is only which way round the integers are written. That the layout really is shared is the corpus's verdict, not an assumption: across 136 DCL-era documents the slots tile the pool exactly and every entry has SETBIT set.

The note record

A note record: six bytes, two of them carrying both pitch and alteration.

```
// byte order: little-endian
struct NoteRec {
    uint16  tcd;                        // +0x0    upper 12 bits pitch (signed), l...
    uint32  flags;                      // +0x2    TIE_START 0x40000000, TIE_END 0...
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	tcd	upper 12 bits pitch (signed), low nibble alteration
	0x02–0x05	4	uint32	flags	TIE_START 0x40000000, TIE_END 0x20000000

Example bytes (6 shown), tinted to match the table above.

```
0000  09 04 00 00 00 40                                    .....@
```

> **The alteration is sign-and-magnitude, not two's complement.** Bit 3 is the sign. `eeppd.txt` describes it as "a signed quantity ... -8 to +7", which reads as two's complement; the corpus disagrees, and the corpus wins. Observed: 0x0 → 0, 0x1 → +1, 0x9 → −1.

Durations

Durations are in **EDU**, where 1024 is a quarter note and 4096 a whole note. Dotted values are the plain value plus half again, so 1536 is a dotted quarter. Across 136 DCL-era corpus documents, 71,801 durations took just 16 distinct values — every one a note value with 0–2 dots, which is itself evidence the field was read correctly.

8. The `.musx` container

A `.musx` is a ZIP archive. Unlike the two `.mus` eras it needs no bespoke container walking — a standard ZIP reader opens it — but its payload member is encrypted.

MEMBER	CONTENTS
`mimetype`	the fixed string `application/vnd.makemusic.notation`
`META-INF/container.xml`	archive manifest
`NotationMetadata.xml`	document metadata
`score.dat`	**the score** — XOR-encrypted, then zlib
`graphics/*`, `presets/*`	embedded artwork and presets

Reading `score.dat`

1. Read the member's bytes from the archive.
2. XOR with the LCG keystream (§3), which resets every 128 KiB.
3. Inflate the result with zlib.
4. The output is **EnigmaXML**, an XML document whose elements carry the same record types the binary formats encode numerically.

> **Treat the archive as hostile.** A ZIP may declare member sizes that do not match its content, repeat member names, or expand enormously. Cap the inflated size, reject duplicate names, and never resolve a member path outside the archive root.

Why one reader covers all three

EnigmaXML names its record types symbolically — `measSpec`, `gfhold`, `entry` — where the 2011 era numbers them and the DCL era uses ETF's two-character tags. These are three spellings of one vocabulary. A reader that converges all three onto a single document model gets the rest of the pipeline for free, which is what this project does: the container differs, the music does not.

CONCEPT	.MUSX	2011 .MUS	DCL .MUS
measure	`measSpec`	176	`^MS`
staff	`staffSpec`	231	`^IS`
frame	`frameSpec`	146	`^FR`
frame hold	`gfhold`	1044	`^GF`
note group	`entry`	entry pool	entry pool

9. Record catalog

Every record type this project decodes, with the offsets its reader actually uses. Fields not listed are present in the payload but not yet established; they are omitted rather than guessed at.

measSpec — tag 176, DCL `^MS`

measSpec — one per measure, keyed by measure number. 2011 tag 176; DCL tag `^MS`.

```
// byte order: little-endian
struct MeasSpec {
    uint16  width;                      // +0x0    measure width in EVPU
    uint16  key;                        // +0x2    key signature; 0 means inherit
    uint16  beats;                      // +0x4    time signature numerator
    uint16  divbeat;                    // +0x6    EDU per beat; 1024 = quarter
    uint16  flags;                      // +0xa    low byte holds the barline nibble
};
```

OFFSET	SIZE	TYPE	FIELD	MEANING
`0x00–0x01`	2	`uint16`	`width`	measure width in EVPU
`0x02–0x03`	2	`uint16`	`key`	key signature; 0 means inherit
`0x04–0x05`	2	`uint16`	`beats`	time signature numerator
`0x06–0x07`	2	`uint16`	`divbeat`	EDU per beat; 1024 = quarter
`0x0a–0x0b`	2	`uint16`	`flags`	low byte holds the barline nibble

Example bytes (12 shown), tinted to match the table above.

```
0000  58 02 03 00 04 00 00 04 00 00 02 00              X...........
```

> **A key of 0 means “inherit”, not C major.** A .musx omits the element entirely rather than writing zero. Emitting `key="0"` would silently turn every inheriting measure into C major.

> **The barline nibble is the low byte of the u16 at +10, not the byte at +10.** ETF stores others data as two-byte values, so on a big-endian file that byte is at +11. Reading +10 either way took the high byte from all 37 big-endian DCL documents.

frameSpec — tag 146, DCL ^FR

frameSpec — a run of entries forming one layer of one measure. 2011 tag 146; DCL tag ^FR.

```
// byte order: little-endian
struct FrameSpec {
    uint8[]  header;                    // +0x0    era-dependent lead-in
    uint32   startEntry;                // +0x6    first entry number in this frame
    uint32   endEntry;                  // +0xa    last entry number, inclusive
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x05	6	uint8[]	header	era-dependent lead-in
	0x06–0x09	4	uint32	startEntry	first entry number in this frame
	0x0a–0x0d	4	uint32	endEntry	last entry number, inclusive

Example bytes (14 shown), tinted to match the table above.

```
0000  00 00 00 00 00 00 65 00 00 00 6c 00 00 00        ......e...l...
```

> **The base offset differs by era: 4 for 2001, 6 for 2005.** The reader distinguishes them by incidence count — a three-incidence spec is the 2001 shape. Reading the wrong base yields entry numbers that look plausible and are wrong, which is the worst kind of error this format offers.

> When a spec has more than one incidence, a `startTime` u32 sits at +0, in EDU from the start of the measure.

gfhold — tag 1044, DCL ^GF

gfhold (“frame hold”) — keyed by two cmpers, (staff, measure). 2011 tag 1044; DCL tag ^GF. This is the record that connects a place in the score to its music.

```
// byte order: little-endian
struct GfHold {
    uint16  clefID;                     // +0x0    clef in force for this measure
    uint16  frame1;                     // +0x6    layer 1 frame id; 0 = layer empty
    uint16  frame2;                     // +0x8    layer 2 frame id
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	clefID	clef in force for this measure
	0x06–0x07	2	uint16	frame1	layer 1 frame id; 0 = layer empty
	0x08–0x09	2	uint16	frame2	layer 2 frame id

Example bytes (10 shown), tinted to match the table above.

```
0000  01 00 00 00 00 00 29 00 00 00                ......)...
```

> A frame of **0 means the layer is empty** and should be omitted, matching a .musx, where an absent slot reads as "this layer holds nothing".

> The format provides four layer slots. Only the first two are decoded here; no corpus document was found carrying a third or fourth.

staffSpec — tag 231, DCL `^IS`

staffSpec — one per staff, keyed by staff number. 2011 tag 231; DCL tag `^IS`. Only the transposition field is decoded.

```
// byte order: little-endian
struct StaffSpec {
    uint16  transposition;              // +0x14   written-to-sounding interval
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x14–0x15	2	uint16	transposition	written-to-sounding interval

Example bytes (22 shown), tinted to match the table above.

```
0000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0010  00 00 00 00 f9 ff                              ......
```

> The staff's *name* is not here — it lives in the text pool, which is why a .mus converts with positional part names unless that pool is resolved.

tupletDef — tag 1072

tupletDef — 2011 tag 1072. The example reads "3 quarters in the time of 2 quarters".

```
// byte order: little-endian
struct TupletDef {
    uint16  symbolicNum;                // +0x0    printed count, the 3 of a triplet
    uint16  symbolicDur;                // +0x2    printed unit in EDU
    uint16  refNum;                     // +0x4    count these replace
    uint16  refDur;                     // +0x6    unit they replace, in EDU
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	symbolicNum	printed count, the 3 of a triplet
	0x02–0x03	2	uint16	symbolicDur	printed unit in EDU
	0x04–0x05	2	uint16	refNum	count these replace
	0x06–0x07	2	uint16	refDur	unit they replace, in EDU

Example bytes (8 shown), tinted to match the table above.

```
0000  03 00 00 04 02 00 00 04                        ........
```

> The sounded duration of an entry inside a tuplet is its written duration scaled by `(refNum × refDur) / (symbolicNum × symbolicDur)`.

staffGroup — tag 1057

staffGroup — the braces and brackets down the left edge. 2011 tag 1057.

```
// byte order: little-endian
struct StaffGroup {
    uint16  startInst;                  // +0x0    first staff in the group
    uint16  endInst;                    // +0x2    last staff, inclusive
    uint16  bracketId;                  // +0xa    bracket shape
    uint8   barlineFlags;               // +0x15   bit 0: barlines cross the group
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	startInst	first staff in the group
	0x02–0x03	2	uint16	endInst	last staff, inclusive
	0x0a–0x0b	2	uint16	bracketId	bracket shape
	0x15	1	uint8	barlineFlags	bit 0: barlines cross the group

Example bytes (24 shown), tinted to match the table above.

```
0000  01 00 04 00 00 00 00 00 00 00 03 00 00 00 00 00  ................
0010  00 00 00 00 00 01 00 00                          ........
```

lyricVerse — tag 1108

lyricVerse — 2011 tag 1108. The payload is an array of 20-byte slots, one per entry that sings.

```
// byte order: little-endian
struct LyricVerseSlot {
    uint16  number;                     // +0x0    verse number
    uint16  syll;                       // +0x2    syllable index into the text pool
    uint16  wext;                       // +0x8    non-zero: word extends with a m...
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	number	verse number
	0x02–0x03	2	uint16	syll	syllable index into the text pool
	0x08–0x09	2	uint16	wext	non-zero: word extends with a melisma

Example bytes (20 shown), tinted to match the table above.

```
0000  01 00 07 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0010  00 00 00 00                                      ....
```

articAssign — tag 1009

articAssign — 2011 tag 1009. Points at an articDef (tag 121), which carries the glyph.

```
// byte order: little-endian
struct ArticAssign {
    uint16  definition;                 // +0x0    key of the articDef this uses
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	definition	key of the articDef this uses

Example bytes (4 shown), tinted to match the table above.

```
0000  0c 00 00 00                                      ....
```

instUsed — tag 159

instUsed — 2011 tag 159. A 24-byte slot per incidence, giving the score's staff order. This, not the staff number, is what a part list should follow.

```
// byte order: little-endian
struct InstUsedSlot {
    uint16  staff;                      // +0x0    staff number, one per incidence
    uint16  staff[1];                   // +0x18   the next slot, 24 bytes on
};
```

	OFFSET	SIZE	TYPE	FIELD	MEANING
	0x00–0x01	2	uint16	staff	staff number, one per incidence
	0x18–0x19	2	uint16	staff[1]	the next slot, 24 bytes on

Example bytes (48 shown), tinted to match the table above.

```
0000  01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0010  00 00 00 00 00 00 00 00 02 00 00 00 00 00 00 00  ................
0020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

Recognized but only partly decoded

These records are read and their keys resolved, but their payload layouts are not fully established. They are listed so an implementer knows they exist rather than meeting them as unexplained bytes.

RECORD	TAG	WHAT IT CARRIES
`articDef`	121	the glyph an articAssign refers to; character offsets 0 and 2
`repeatBack`	203	backward repeat barline; target measure
`repeatEndingStart`	204	first and second ending brackets
`repeatPassList`	206	which passes an ending applies to
`clefOptions`	109	the clef definition table, read as a strided array

References

References

1. *Enigma Transportable File Specification*, Coda Music Technologies. Identified by the Library of Congress as version 98c.0, for Finale 97 for Mac and Windows: `loc.gov/preservation/digital/formats/fdd/fdd000633.shtml`. Vendored as `docs/etfspec.pdf`.
2. *Enigma Entry Pool: preliminary documentation*, Coda Music Technologies, 8 February 1996. Archived at `web.archive.org/web/19990203113959/http://www.codamusic.com:80/down/finale/eeppd.txt`. Vendored as `docs/eeppd.txt`.
3. Principally *denigma* (MIT), `github.com/chrisroode/denigma`, which credits Deguerre for the `score.dat` keystream; the *EnigmaXML documentation* (MIT), `github.com/Project-Attacca/enigmaxml-documentation`; and *musxdom* (MIT), `github.com/rpatters1/musxdom`. The full list, with what each contributed, is in `docs/REFERENCES.md`.
4. *Enigma Entry Pool: preliminary documentation* (see note 2), which records the term: “Certain situations like mirrors and voice 2 create complications.”